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ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-2** **Date: 27-03-2026**

Time: 03.00PM TO 06.00PM **GTM-30** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature: _____**

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CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. An incident ray is reflected by the line mirror y = 1 at the point (2,**1**). If the reflected ray touches the circle $x^2+y^2=1$. Then the equation of incident ray is

1) 3x -4y -2 = 0 **2)** 3x-4y-10=0 **3)** 4x-3y-5=0 **4)** 4x+3y-11=0

2. Locus of the feet of the perpendiculars drawn from vertex of the parabola $y^2=4ax$ upon all such chords of the parabola which subtend a right angle at the vertex is…

1) $x^2+y^2-4ax=0$ **2)** $x^2+y^2-2ax=0$ **3)** $x^2+y^2+2ax=0$ **4)** $x^2+y^2+4ax=0$

3. For each positive integer n, consider the point P with abscissa n on curve $y^2-x^2=1$ if d_n represents the shortest distance from the point P to the line y = x then $\lim_{n\to\infty}(n.\,d_n)$ has value equal to

1) $\frac{1}{2\sqrt{2}}$ **2)** $\frac{1}{2}$ **3)** $\frac{1}{\sqrt{2}}$ **4)** 0

4. Let A, B, C, D be (not necessarily square) real matrices such that $A^T=BCD; B^T=CDA; C^T=DAB$ and $D^T=ABC$ for the matrix S = ABCD, consider the two statements

I : $S^3=S$

II : $S^2=S^4$

1) II is true but not I **2)** I is true but not II

3) Both I and II are true **4)** both I and II are false

5. Which of the following is/are FALSE?

1) A and B are two independent events. If the probability that both A and B occur is $\frac{1}{12}$ and the probability that neither A nor B occur is $\frac{1}{2}$, then $P(A-B)=\frac{1}{6}$. (given P (A) < P(B))

2) A fair coin is Tossed repeatedly. If the head appears on the first three tosses, then the probability that the tail appearing on the fourth toss equals $\frac{1}{2}$.

3) If the letters of the word "FREEDOM" are written down at random in a row, the probability that no two E's occur next to each other is $\frac{6}{7}$

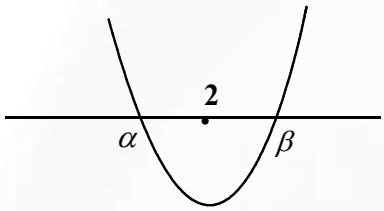
4) For two give events A and B, P(A)+P(B)-1 $\le P(A\cap B)\le P(A)+P(B)$

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6. If $f(x)=x^3+3x+4$ and g is inverse function of f then the value of $\frac{d}{dx}\left(\frac{g(x)}{g(g(x))}\right)$ at x = 4 equals

1) $\frac{-1}{6}$ **2)** 6 **3)** $\frac{-1}{3}$ **4)** non-existent

7. Let α and β be the roots of the equation $x^2+2ax+(3a+10)=0$ such that $\alpha<2<\beta$. Then the set of all possible values of a is…..



1) $(-\infty\ -2)$ **2)** $(-\infty\ 5)$ **3)** $(-\infty\ -2)\cup(5\ \infty)$ **4)** $(5\ \infty)$

8.

	Column-I		Column-II
A	Number of triangle that can be made using the vertices of a polygon of I0 sides as their vertices and having exactly one side common with the polygon is	P	75
B	Number of triangle that can be made using the vertices of a polygon of 10 sides as their vertices and having exactly 2 sides common with the polygon is	Q	110
C	Number of quadrilaterals that can be made using the vertices of a polygon of 10 sides as their vertices and having exactly 2 sides common with the polygon is	R	60
D	Number of quadrilaterals that can be made using the vertices of a polygon of 10 sides as their vertices bad having 3 sides common with the polygon is	S	10

1) A-R, B-S, C-P, D-Q **2)** A-S, B-R, C-P, D-S

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3) A-R, B-S, C-P, D-S **4)** A-S, B-R, C-P, D-Q

9. Let f be a positive continuous on the interval [-2,3] and A (t) is the area of the region bounded by the graph of $y=f(x)$ and the lines $y=0, x=-2$ and $x=t$ where $t\in(-2,3)$. If $\lim\limits_{t\to3^-}\frac{A(3)-A(t)}{3-t}=100$ then the value of $f(3)$ equals

1) 3 **2)** 100 **3)** $\frac{1}{100}$ **4)** 300

10. A curve passes through (2,0) and the slope of tangent at point P (x, y) equals $\frac{(x+1)^2+y-3}{(x+1)}$. The area enclosed by the curve and the x-axis in the fourth quadrant will be

1) $\frac{1}{3}$ **2)** $\frac{2}{3}$ **3)** 1 **4)** $\frac{4}{3}$

11. Let A = {2, 3, 5, 7, 9}let R be the relation on A defined by x R y if and only If $2x\le3y+1$. Let 'l' be the number of elements in R and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then l + m is

1) 25 **2)** 24 **3)** 26 **4)** 23

12. A complex number z is said to be unimodular if $|z|$ =1. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1-2z_2}{2-z_1\bar{z}_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a

1) circle of radius 2 **2)** circle of radius $\sqrt{2}$

3) straight line parallel to x-axis **4)** straight line parallel to y-axis

13. If the system of linear equations

$3x+y+\beta z=3$,

$2x+\alpha y-z=-3$,

$x+2y+z=4$,

Has infinitely many solutions, then the value of $22\beta-9\alpha$ is

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1) 49 **2)** 43 **3)** 31 **4)** 37

14. If the function

$$\begin{cases} \frac{2}{x}\{\sin(k_1+1)x+\sin(k_2-1)x\}, & x<0 \\ 4, & x=0 \\ \frac{2}{x}\log_e\left(\frac{2+k_1x}{2+k_2x}\right), & x>0 \end{cases}$$

Is continuous at x = 0, then $k_1^2+k_2^2$ is equal to

1) 20 **2)** 5 **3)** 10 **4)** 8

15. Let $\bar{a}$ and $\bar{b}$ be two vectors such that $|\bar{b}|=1$ and $|\vec{b}\times\vec{a}|=2$. Then $\left(\left(|\vec{b}\times\vec{a}|-\vec{b}\right)\right)^2$ is equal to

1) 3 **2)** 5 **3)** 1 **4)** 4

16. Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}, -5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocenter and the circumference of the triangle are $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ and $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ respectively, then $\alpha+2\beta+5\gamma$ is equal to…

1) 1 **2)** 3 **3)** 6 **4)** 4

17. The shortest distance, between lines L_1 and L_2,where $L_1: \frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ and L_2 is the line, passing through the points A(-4,4,**3**), B(-1,6,**3**) and perpendicular to the line $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{2}$, is

1) $\frac{42}{\sqrt{117}}$ **2)** $\frac{121}{\sqrt{221}}$ **3)** $\frac{24}{\sqrt{117}}$ **4)** $\frac{141}{\sqrt{221}}$

18. Let the range of the function

$f(x)=6+16\cos x.\cos\left(\frac{\pi}{3}-x\right)\cdot\cos\left(\frac{\pi}{3}+x\right)\cdot\sin 3x\cdot\cos 6x, x\in R$ be $[\alpha,\beta]$. Then the distance of the point (α,β) from the line $3x-4y-12=0$ is.

1) 8 **2)**5 **3)** 9 **4)** 10

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19. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{[x]+5}\right)dx$. where [.] denotes greatest integer functions

1) $\frac{\pi}{4}+\frac{1}{20}$ **2)** $\frac{\pi}{4}-\frac{1}{20}$ **3)** $\frac{5\pi+1}{10}$ **4)** $\frac{\pi}{4}-\frac{1}{10}$

20. Let $x=\frac{m}{n}$ (m, n are co- prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right)=\frac{1}{9}$ and let $\alpha,\beta(a>\beta)$ be the roots of the equation $mx^2-nx-m+n=0$. Then the point (α,β) lines on the line

1) $3x+2y=2$ **2)** $5x-8y=-9$ **3)** $3x-2y=-2$ **4)** $5x+8y=9$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. The parallelogram is bounded by the lines $y=ax+c; y=ax+d; y=bx+c$ & $y=bx+d$ has the area equal to 18. The parallelogram bounded by the lines $y=ax+c; y=ax-d; y=bx+c$ and $y=bx-d$ has area 72. Given that a, b, c and d are positive integers, find the smallest possible value of (a + b +c + d)

22. Consider a function of the form $f(x)=\alpha e^{2x}+\beta e^{x}-\gamma x$ where α,β,γ are independent of x and $f(x)$ satisfies the following conditions $f(0)$ = -1 $f'(\ell n2)=30$ and $\int_0^{\ell n4}\left(f(x)+\gamma x\right)dx=24$ find the value of $(\alpha+\beta+\gamma)$.

23. If the coefficients so x^{r-2}, x^{r-1} and x^{r} of $(1+x)^{n+5}$ are in the ratio 5:10:14 then 5n =

24. Let 729, 81,9,1, ……..be a sequence and p_n denotes the product of the first 'n' terms of the sequence. If $2\sum_{n=1}^{50}(p_n)^{\frac{1}{n}}=\frac{3^{\alpha}-1}{3^{\beta}}$, then $\alpha+\beta=$

25. Let the sets of all values of p, for which $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ does not have any critical point, be the interval (a,b). Then 16 ab is equal to….

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. The transverse displacement of a string (clamped as its two ends) is given by $y(x,t)=0.06\sin\left(\frac{2\pi}{3}x\right)\cos(120\pi t)$ where x,y are in m and t in s. The length of the string is 1.5m and its mass is 3×10^{-2} kg.

Determine tension in string

1) 628N **2)** 638 N **3)** 648 N **4)** 658 N

27. Answer the following questions:

Statement-I A diver under water, looks obliquely at a fisherman standing on the bank of a lake. the fisherman look taller to the diver than what he actually is

Statement –II the apparent depth of a tank of water change if viewed obliquely If so, the apparent depth decrease

1) Statement I is true statement II false

2) Statement I is false statement II false

3) Statement I is true statement II true

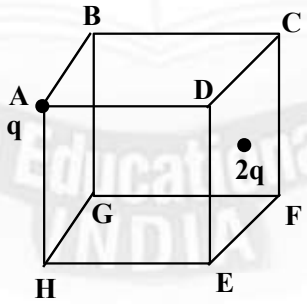
4) Statement I is false statement II true

28. Monochromatic light of wavelength 589 nm is incident from air on a water surface. What is the frequency of reflected light

1) $f=5.00\times10^{14}\,Hz$ **2)** $f=5.09\times10^{14}\,Hz$

3) $f=5.09\times10^{15}\,Hz$ **4)** $f=5.09\times10^{13}\,Hz$

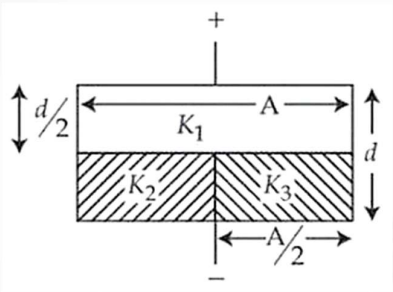
29. Two points charges q and 2q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing through the cube is



1) $\frac{3q}{2\varepsilon_0}$ **2)** $\frac{q}{\varepsilon_0}$ **3)** $\frac{9q}{8\varepsilon_0}$ **4)** $\frac{3q}{4\varepsilon_0}$

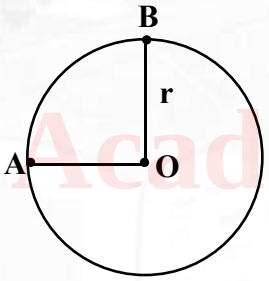
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30. The space between the plates of a parallel plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2, K_2 = 3$ and $K_3 = 5$ (as shown in figure) If new capacitance is $\frac{n}{3}C$ then the value of n is…….



1) 7 **2)** 8 **3)** 9 **4)** 10

31. A wire of uniform resistance $\lambda\Omega/m$ is bent into a circle of radius r and another piece of wire with length 2r is connected between points A and B(AOB) as shown in figure. The equivalent resistance between points A and B is…Ω



1) $\frac{6\pi\lambda r}{3\pi+16}$ **2)** $(\pi+1)2r\lambda$ **3)** $2\pi\lambda r$ **4)** $\frac{3\pi\lambda r}{8}$

32. A horizontal straight wire 10 m long extending from east to west is falling with a speed of 5.0 m/s, at right angles to the horizontal component of the earth's magnetic field, 0.60×10^{-4} Wb/m^2.

i) What is the instantaneous value of the emf induced in the wire?

ii) Which end of the wire is at the higher electrical potential?

1) $3\times10^{-3}V$ east **2)** $3\times10^{-4}V$ west

3) $1.5\times10^{-5}V$ east **4)** $1.5\times10^{-3}V$ west

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33. Which of the following statement are false:

a) In an elastic collision of two bodies, the momentum and energy of each body is conserved.

b) Total energy of a system is always conserved, no matter what internal and external forces on the body are present.

c) Work done in the motion of a body over a closed loop is zero for every force in nature.

d) In an inelastic collision, the final kinetic energy is always less than the initial kinetic energy of the system.

1) a,b only **2)** a,c only **3)** a,c,d only **4)** a,b,c only

34. The ratio of speed of electromagnetic waves in vacuum and a medium is $\sqrt{6}$ having dielectric Constant k = 3 and permeability of $\mu = \ldots\ldots$ (μ_0 = permeability of vacuum)

1) μ_0 **2)** $2\mu_0$ **3)** $3\mu_0$ **4)** $6\mu_0$

35. The second member of Lyman series in hydrogen spectrum has wavelength 5400 A^0. Find the wavelength of first member.

1) 6400 A^0 **2)** 3200 A^0 **3)** 4800 A^0 **4)** 1600 A^0

36. The resistance connected in series to the galvanometer is greater because

Assertion -A Voltmeter has to draw a negligible amount of current from the main circuit, so resistance of the voltmeter should be high.

Reason -R: To increase the effective resistance of the voltmeter, the resistance connected in series should be greater than the galvanometer resistance

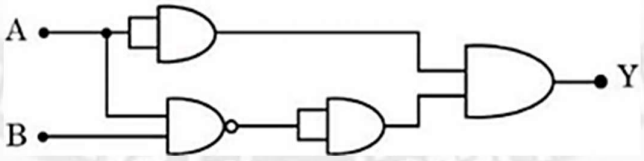
1) A and R both are correct and R is correct explanation of A

2) Both A and R correct but R is not the correct explanation of A

3) A is correct but R is not correct

4) A is not correct but R is correct

37. Identify the correct table of the of the given login circuit.



1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

2)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

3)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

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38. **Statement-I**: To keep a piece of paper horizontal, you should blow over, not under it

Statement II: A spinning cricket ball in air does not follow a parabolic trajectory.

1) Both statement I and Statement II are true

2) Both statement I and Statement II are false

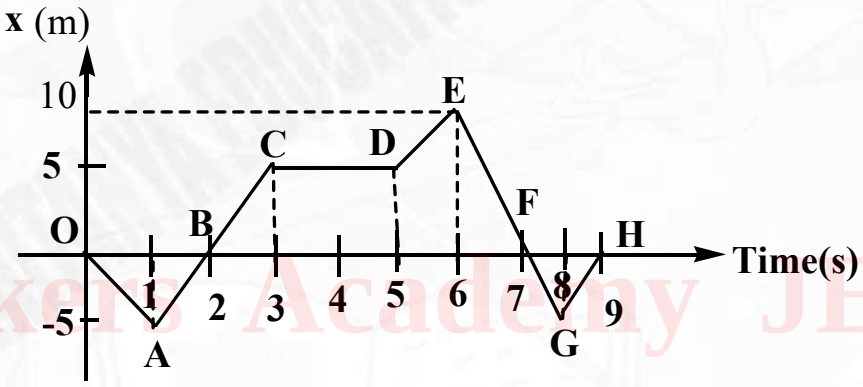
3) Statement I is true but Statement Il are false

4) Statement I is false but Statement II is true

39. A quantity $f = \sqrt{\frac{hc^2}{G}}$ where c is speed of light, G gravitational constant and h is planks constant, L is length then dimensions of f is…

1) Energy **2)** Force **3)** volume **4)** area

40. The displacement x versus time graph is shown below



Column-I		Column-II	
A	The average velocity during 0 to 3 sec is	P	10 m/s
B	The instantaneous velocity at t = 4 is	Q	5/3 m/s
C	The average velocity during t = 2 to t = 6sec is	R	zero
D	Acceleration at t = 3 sec to t = 5 sec is	S	2.5 m/s

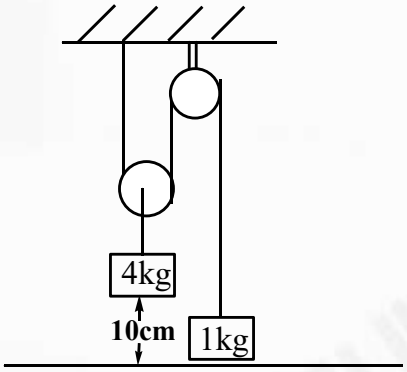
Choose correct option

1) A-R, B-Q, C-S, D-R **2)** A-Q, B-R, C-R, D-P

3) A-Q, B-R, C-S, D-R **4)** A-Q, B-S, C-R, D-R

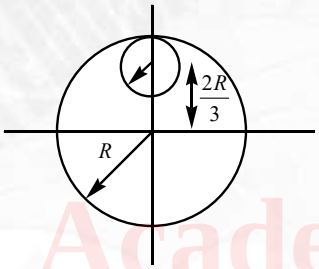
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41. The boxes of masses 1kg and 4 kg are connected by massless string passing over a smooth pulleys as shown. Calculate the time taken by box of mass 4 kg to strike the ground starting from rest



1) 0.45 **2)** 0.2 sec **3)** 0.25 sec **4)** 0.35 sec

42. From a uniform circular disc of radius R and mass 4 M, a small disc of radius R/2 is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is:



1) $\frac{37}{8}MR^2$ **2)** $4MR^2$ **3)** $\frac{13}{8}MR^2$ **4)** $10MR^2$

43. Given below are two statements:

Statement I: The law of gravitation holds good for any pair of bodies in the universe.

Statement II: The weight of any person becomes zero when the person is at the centre of the earth. In the light of the above statements, choose the correct answer from the options given below.

1) Both statement I and Statement II are true

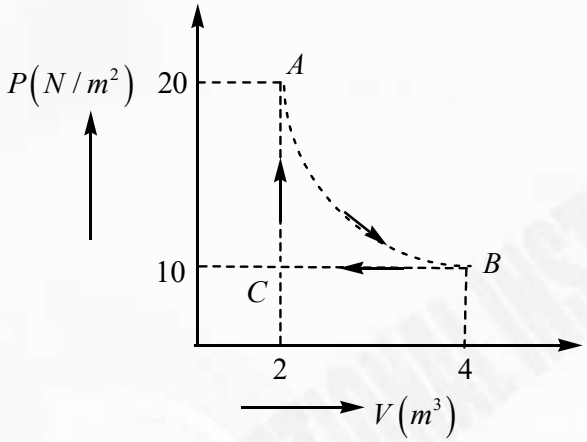
2) Both statement I and Statement II are false

3) Statement I is true but Statement Il are false

4) Statement I is false but Statement II is true

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44. A real gas within a closed chamber at 27°C undergoes the cyclic process as shown in figure. The gas obeys $PV^2 = RT$ equation for the path A to B. The net work done in the complete cycle is (assuming R = 8 J/ mol K)



1) 600 J **2)** 205 J **3)** 20 J **4)** -20 J

45. The magnetic field of a plane electromagnetic wave is given by $\vec{B} = 3\times10^{-8}\sin\left(0.5\times10^{3}x + 1.5\times10^{11}t\right)\hat{j}T$ The amplitude of the electric field would be

1) $9Vm^{-1}$ along x-axis **2)** $3Vm^{-1}$ along z-axis

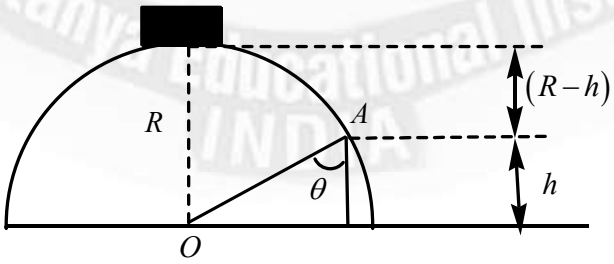
3) $9Vm^{-1}$ along z-axis **4)** $6\times10^{-8}Vm^{-1}$ along z-axis

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

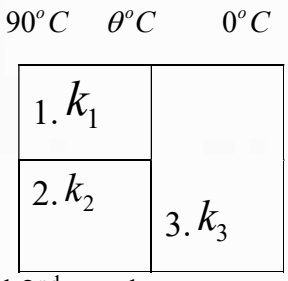
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A small block slides down from the top of hemisphere of radius R=6 m as shown in the figure. The height 'h' at which the block will lose contract with the surface of the sphere is …..m



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47. Three conductors of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in figure.



Area of cross sections of 1st and 2nd conductors are same and for 3rd conductors it is double of the 1st conductor. The temperatures are given in the figure. In steady state condition, the value of θ is …….°C.

(given: $k_1 = 60\,Js^{-1}m^{-1}, K^{-1} = 120\,Js^{-1}m^{-1}K^{-1}, k_3 = 180\,Js^{-1}m^{-1}K^{-1}$)

48. A tiny metallic rectangular sheet has length and breadth of 6 mm and 1.5 mm, respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 25 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{1000}$ where x is ……

49. A closely wound solenoid 80 cm long has N layers of winding of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0A, the magnitude of B inside the solenoid near its centre is 2.5×10^{-2} T find N.

50. A light hollow cube of side length 10 cm and mass 40 g, is floating in water. It is pushed down and released to execute simple harmonic oscillations. The time period of oscillations is $y\pi\times10^{-2}$ s, where the value of y is (Acceleration due to gravity, g = 10 m/s ^{2}, density of water = 10^3 kg/m^3)

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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. 400 cm^3 of an aqueous solution of protein contains 1.26 gm of the protein. The osmotic pressure of such a solution at 300K is found to be 2.57×10^{-3} bar. The Molar mass of the protein is …..(R = 0.083 Lt bar/K.mol)

1) 30519 gm **2)** 29650 gm **3)** 38000 gm **4)** 32525 gm

52. Given below are two statements:

Statement-I: The metallic radius of Na is 1.86 A^o and the ionic radius of Na^+ is greater than 1.86 A^o.

Statement-II: ions are always greater in size than the corresponding elements.

In the light above statements, choose the correct answer from the options given below.

1) Both statements I and II are true

2) Statement I is incorrect but Statement II is true

3) Both statement I and statement II are false

4) Statement I correct but Statement II is false

53. Match list I with List II

	List –I (Molecule)		**List –II (Hybridization, shape)**
A	XeO_3	I	sp^3d^2, square pyramidal
B	SF_4	II	dsp^2, square planar
C	BrF_5	III	sp^3, pyramidal
D	$[PtCl_4]^{-2}$	IV	sp^3d, see saw

1) A-I, B-III, C-IV, D-I **2)** A-III, B-IV, C-I, D-II

3) A-III, B-II, C-I, D-IV **4)** A-II, B-III, C-III, D-I

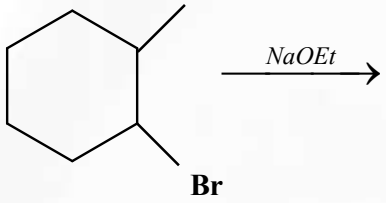
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JEE Advanced 2023 RANK 1
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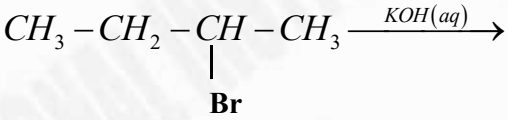
54. One of the reaction that takes place in producing steel from iron is the reduction of iron (II) Oxide by Carbon monoxide to give iron metal and CO_2.

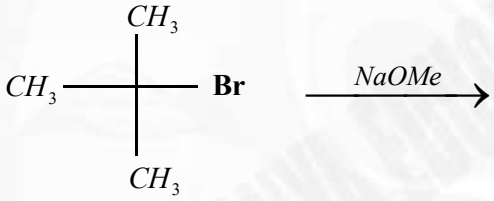
$FeO_{(S)} + CO_{(g)} \rightleftharpoons Fe_{(s)} + CO_{2(g)}$; $K_P = 0.465\,atm$ at 1050K. What is the equilibrium partial pressure of CO_2. At 1050K if the initial partial pressures are P_{CO} = 1.4 atm and P_{CO_2} = 0.8 atm?

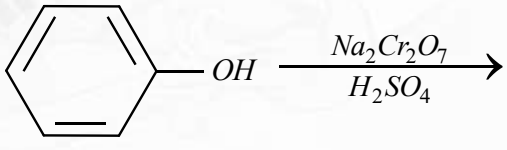
1) 0.443 **2)** 0.699 **3)** 0.892 **4)** 0.432

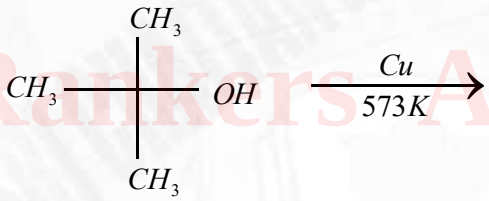
55. The reactions which cannot be applied to prepare an alkene by elimination, are

A) 

B) 

C) 

D) 

E) 

Choose the correct answer from the options give below:

1) B & E only **2)** B, C & D only **3)** A, C & D only **4)** B & D only

56. Identify the correct statements about 15th group elements

A) Dinitrogen is a diatomic gas which acts like an inert gas at room temperature

B) The common oxidation states of these elements are -3, +3 and +5

C) Nitrogen has unique ability to form $d\pi$ - $d\pi$ multiple bonds

D) The stability of +3 oxidation states increases down the group

E) Nitrogen shows a maximum covalency of 6

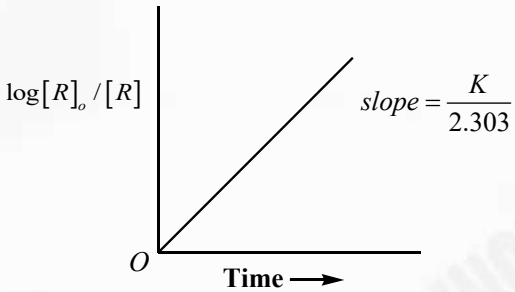
1) A, B, C only **2)** C,D,E only **3)** A,B,D only **4)** B,C,D only

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57. Given below are two statements

Statement –I: The rate law for the reaction A+B→C is, rate $(r)=k[A]^2[B]^o$. When the concentration of both A and B is doubled, the reaction rate is increased x times.

Statement-II:



The figure is showing the variation $\log[R]_o/[R]$ against time plot for a 'y' order reaction.

The value of x + y is..

1) 4 **2)** 3 **3)** 5 **4)** 6

58. Given below are two statements: one is labelled as assertion A and the other is labelled as reason R.

Assertion A: Gabriel phthalimide synthesis can be used to prepare aromatic primary amines

Reason R: aryl halides do not readily undergo nucleophilic substitution reaction

In the light of the above statements, choose the correct answer from the options given below.

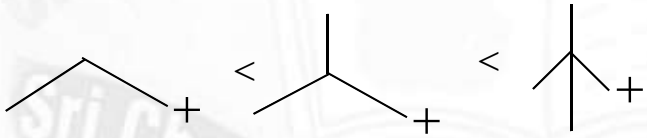
1) Both A and R are true and R is the correct explanation of A.

2) Both A and R are true but R is not the correct explanation of A

3) A is true but R is false

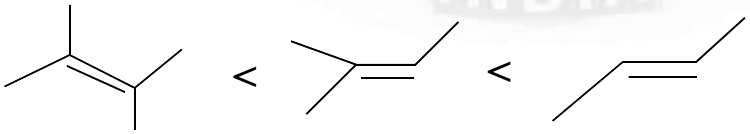
4) A is false but R is true

59. Which of the following is correct order

1)  **(Stability of carbocation)**

2)  **(B.P order)**

3) $Ph-CH=CH-CH_2-CH_3 < Ph-CH_2-CH=CH-CH_3 < Ph-CH_2-CH_2-CH=CH_2$ (order of heat of hydrogenation)

4)  (Stability of alkene)

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60. Identify correct statements from below

A) The chromate ion is tetrahedral

B) Sodium dichromate is less soluble than potassium dichromate

C) The green manganate ion is paramagnetic but the permanganate ion is diamagnetic.

D) Acidified permanganate solution oxidizes oxalate to carbon dioxide.

E) In vanadium there is gradual change from the less basic V_2O_3 to more basic V_2O_4 and to amphoteric V_2O_5. Choose the correct answer from the options given

1) A, C, D **2)** A, D, E **3)** B, C, D, E **4)** A, B, C,E

61. Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according the reaction.

$4HCl_{(aq)} + MnO_{2(s)} \rightarrow 2H_2O_{(l)} + MnCl_{2(aq)} + Cl_{2(g)}$ the mass of HCl required to react with 50 gm of manganese dioxide is….

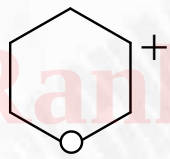
1) $8.4 gms$ **2)** $84 gms$ **3)** $9.4\ gms$ **4)** $94\ gms$

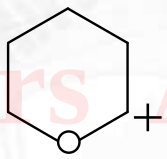
62. The number of species from the following that are involved in d^2sp^3 hybridization is

$\left[Co(NH_3)_6\right]^{+3}, \left[CrF_6\right]^{-3}, \left[CoF_6\right]^{-3}, \left[Fe(CN)_6\right]^{-3}, SF_6, PCl_6^-$

1) 5 **2)** 2 **3)** 4 **4)** 3

63. Order of stability of following carbocations


A)


B)


C)

1) C > B > A **2)** A > B > C **3)** B > C > A **4)** C > A > B

64. Given below are two statements: one is labelled as assertion A and the other is labelled as Reason R

Assertion A: Amylose is water soluble component which constitutes about 15-20 % of starch

Reason R: Amylopectin is water soluble component and constitutes about 80-85% of starch.

In the light of the above statements, choose the correct answer from the options given below

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A

3) A is true but R is false

4) A is false but R is true

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65. The difference between the radii of 3rd and 4th orbits of Li^{+2} is ΔR_1. The difference between the radii of 3rd and 4th orbits of He^{+} is ΔR_2 then ratio of $\Delta R_1 : \Delta R_2$ is……

1) 8: 3 **2)** 3 : 8 **3)** 2 :3 **4)** 3:2

66. The correct statements from the following are

A) The decreasing order of atomic radii of group 13 elements is $Tl > In > Ga$

B) Down the group 13 elements electronegative decreases from top to bottom

C) Al dissolve dil HCl and liberate H_2 but con HNO_3 renders Al passive by forming a protective oxide layer on the surface

D) All elements of group 13 exhibits highly stable +3 oxidation state.

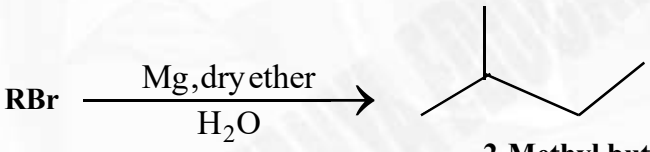
E) Hybridization of Al in $\left[Al(H_2O)_6\right]^{+3}$ ion is sp^3d^2

Choose the correct answer from the options given below

1) A, B, C & E only **2)** A, C & E only

3) C& E only **4)** A & C only

67.

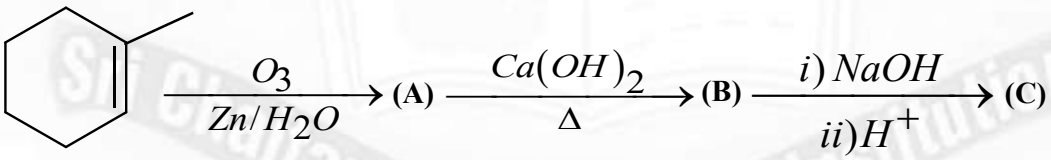


2-Methyl butane

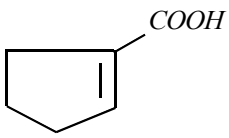
The maximum number of RBr producing 2- methyl butane by above sequence of reactions is…. (Including stereo)

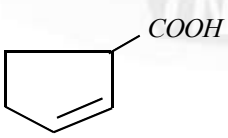
1) 5 **2)** 3 **3)** 6 **4)** 4

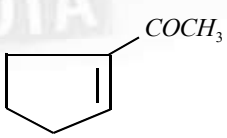
68.

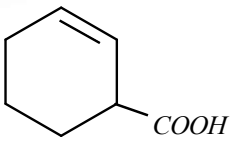


Identify product (C)

1) 

2) 

3) 

4) 

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JEE Advanced 2023 VAVILALA CHAITANYA REDDY RANK 1
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69. How many of the following are in the correct order with respect to the given property?

i) He > Ne > Ar > Kr > Xe: (B.P order)

ii) S > O > Se > Te (electron gain enthalpy with negative sign)

iii) HCl < HBr < HF < HI (acidic order)

iv) I > Cl > Br (stability of oxides of Halogens)

v) MF > MCl > MBr > MI (ionic character of halides)

1) 5 **2)** 4 **3)** 3 **4)** 2

70. The following data were obtained during the first order thermal decomposition of $N_2O_{5(g)}$ at constant volume. (log 1.05 = 0.0212)

$$2N_2O_{5(g)} \rightarrow 2N_2O_{4(g)} + O_{2(g)}$$

S.no	Time /s	Total pressure / (atm)
1	0	0.5
2	150	0.512

Calculate the rate constant

1) $3.25\times10^{-4}\ sec^{-1}$ **2)** $3.25\times10^{-3}\ sec^{-1}$ **3)** $32.5\times10^{-1}\ sec^{-1}$ **4)** $3.05\times10^{-1}\ sec^{-1}$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The volume (in ml) of 0.01 M $AgNO_3$ required for complete precipitation of chloride ions present in 20 ml of 0.1 M solution of $\left[Cr(H_2O)_5Cl\right]Cl_2$ as silver chloride is……..

72. Identify the structure of the final product (D) in the following sequence of the reactions:

$$Ph-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[\Delta]{PCl_5} A \xrightarrow{3\ eq.NaNH_2/NH_3} B \xrightarrow{\text{Acidify}} C \xrightarrow[2.H_2O_2/OH^-]{1.B_2H_6} D$$

Total number of sp^3 hybridized carbon atoms in product D is…….

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
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73. Consider the following half cell reaction

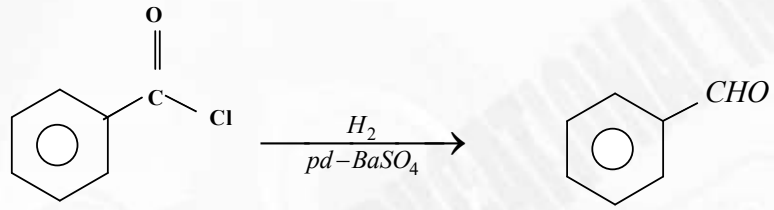
$$Cr_2O_7^{-2}{}_{(aq)} + 6e^- + 14H^+{}_{(aq)} \rightarrow 2\,Cr^{+3}{}_{(aq)} + 7H_2O_{(l)}$$

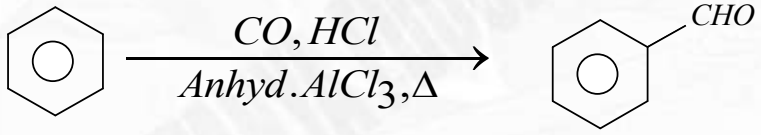
The reaction was conducted with the ratio of $\frac{\left[Cr^{+3}\right]^2}{\left[Cr_2O_7^{-2}\right]} = 10^{-8}$

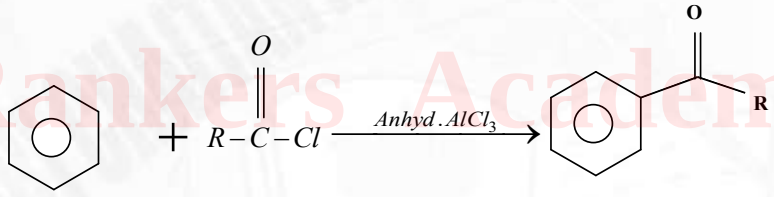
The P^H value at which the EMF of the cell will become zero is ….(nearest integer value)

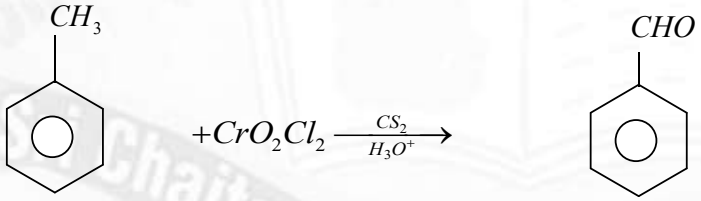
(Given : Standard half cell reduction potential $E^o_{Cr_2O_7^{-2}/Cr^{+3}} = 1.33V$ & $\frac{2.303RT}{F} = 0.059V$)

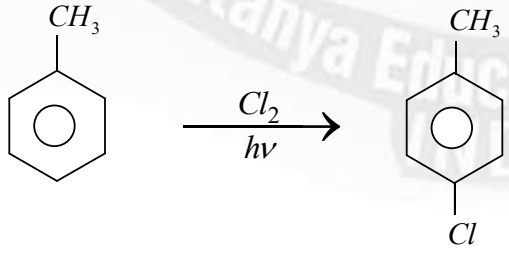
74. The number of correct reaction (s) among the following is……

A) 

B) 

C) 

D) 

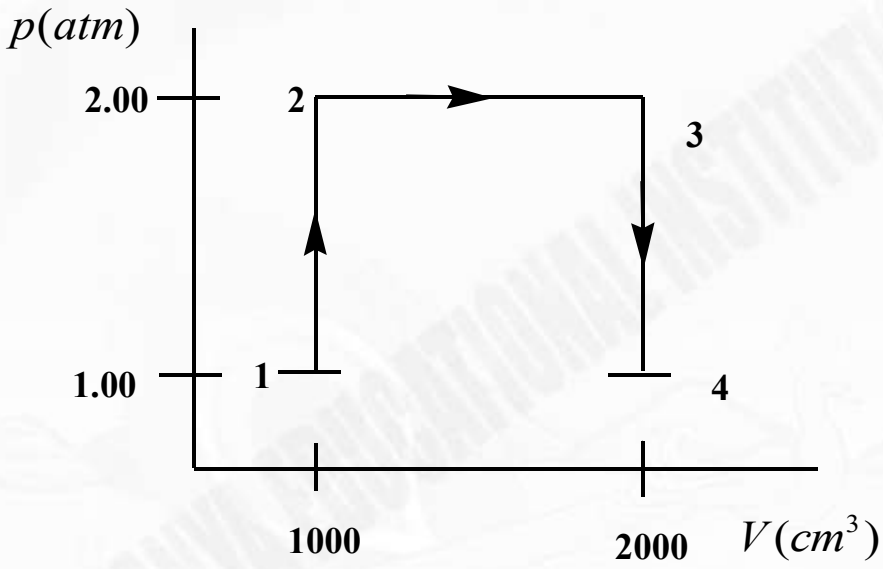
E) 

F) $RMgX + CO_2 \xrightarrow[H_3O^+]{Dry\ ether} RCHO$

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75. A perfect gas (0.1 mol) having $C_V = 1.50R$ (independent of temperature) undergoes the below transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is (-).....J

(Given R = 0.082 L atm K^{-1} mol^{-1})



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